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$$1^2 \cdot 4 + 2^2 \cdot 5 + \dots + n^2(n+3)$$

$$= s_3(n) + 3 \cdot s_2(n)$$

waarbij:

$$s_3(n) = \sum_{k=1}^n k^2 \cdot k = \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2 = \frac{n^2(n+1)^2}{4}$$
$$s_2(n) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Dus:

$$\begin{aligned} &= \frac{n^2(n+1)^2}{4} + 3 \cdot \frac{n(n+1)(2n+1)}{6} \\ &= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} + 3 \cdot \left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) \\ &= \frac{n^4}{4} + \frac{3 \cdot n^3}{2} + \frac{7 \cdot n^2}{4} + \frac{n}{2} \\ &= \frac{1}{4}n(n+1)(n^2+5n+2) \end{aligned}$$

Omdat n of $n+1$ even is en n^2+5n+2 altijd even is, is de teller dus deelbaar door 4.

References